

High-precision, automated stable isotope analysis of atmospheric methane and carbon dioxide using continuous-flow isotope-ratio mass spectrometry

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Small-scale developments have been made to an off-the-shelf continuous-flow gas chromatography/isotope-ratio mass spectrometry (CF-GC/IRMS) system to allow high-precision isotopic analysis of methane (CH₄) and carbon dioxide (CO₂) at ambient concentrations. The repeatability (1 σ) obtainable with this system is 0.05‰ for $\delta^{13}\text{C}$ of CH₄, 0.03‰ for $\delta^{13}\text{C}$ of CO₂, and 0.05‰ for $\delta^{18}\text{O}$ of CO₂ for ten consecutive analyses of a standard tank. An automated inlet system, which allows diurnal studies of CO₂ and CH₄ isotopes, is also described. The improved precision for CH₄ analysis was achieved with the use of a palladium powder on quartz wool catalyst in the combustion furnace, which increased the efficiency of oxidation of CH₄ to CO₂. The automated inlet further improved the precision for both CH₄ and CO₂ analysis by keeping the routine constant. The method described provides a fast turn-around in samples, with accurate, reproducible results, and would allow a long-term continuous record of CH₄ or CO₂ isotopes at a site to be made, providing information about changing sources of the gases both seasonally and interannually. Copyright © 2005 John Wiley & Sons, Ltd.

Atmospheric methane (CH₄) has more than doubled since pre-industrial times^{1,2} due to an imbalance between increasing sources of the gas and its sinks. The 2001 IPCC Climate Change report recognised that the source strengths of CH₄ emissions are largely uncertain due to difficulties in measuring global emission rates of variable biospheric sources.³ Stable isotopic analysis of CH₄ is important in identifying sources and sinks of the gas, and can aid in quantification of the magnitude of emissions from individual sources. Atmospheric CH₄ has a global mean carbon isotopic composition ($\delta^{13}\text{C}$) of -47.3‰ .⁴ However, this value varies both temporally and spatially.

The stable C isotope ratio of atmospheric CH₄ represents a balance between fractionation during production of CH₄ and fractionation by CH₄ sinks. Many sources of CH₄ have distinct isotopic signatures. In general, CH₄ from biogenic sources, e.g. wetlands and landfill sites, is isotopically depleted in $\delta^{13}\text{C}$ ($<-50\text{‰}$), whereas CH₄ produced by pyrogenic and thermogenic sources is more enriched.^{5,6} The isotopic composition of individual sources can vary depending on local conditions, but datasets of $\delta^{13}\text{C}$ values are available for the main sources of CH₄.^{5,7} The largest sink for

CH₄ in the atmosphere is through oxidation by OH radicals, which preferentially oxidise $^{12}\text{CH}_4$.

In the same way, the isotopic composition of atmospheric carbon dioxide (CO₂) provides information on its sources and sinks. Fossil fuel burning, the main anthropogenic source of CO₂, adds CO₂ to the atmosphere that is more depleted than background levels,⁸ averaging -28‰ ⁹ compared to a globally averaged background $\delta^{13}\text{C}$ of -8.2‰ .¹⁰ $\delta^{13}\text{C}$ studies of atmospheric CO₂ have been used to constrain the partitioning of CO₂ uptake by oceans and land. Land uptake by photosynthesis leads to a fractionation of the order of 20‰ (for C₃ plants),¹¹ whereas exchange of CO₂ with the oceans has a fractionation of 2‰ or less.¹²

Isotopic measurements of CH₄ can be used to identify and quantify sources of the gas on global, regional and local scales. On a global scale, measurements at background sites are particularly important. Background sites are located in areas with well-mixed air, often in marine or coastal locations, far from local sources. Background CH₄ mixing ratio and $\delta^{13}\text{C}$ values vary following an annual cycle, due to seasonal changes in OH and to seasonality of some sources. Background CH₄ mixing ratios and $\delta^{13}\text{C}$ values also vary latitudinally, with lowest concentrations and highest $\delta^{13}\text{C}$ in the southern hemisphere reflecting the higher density of sources in the northern hemisphere, especially isotopically depleted biogenic sources such as wetlands. The amplitude of the seasonal cycle is smallest at low southern latitudes where seasonal variations in source strengths are lowest.

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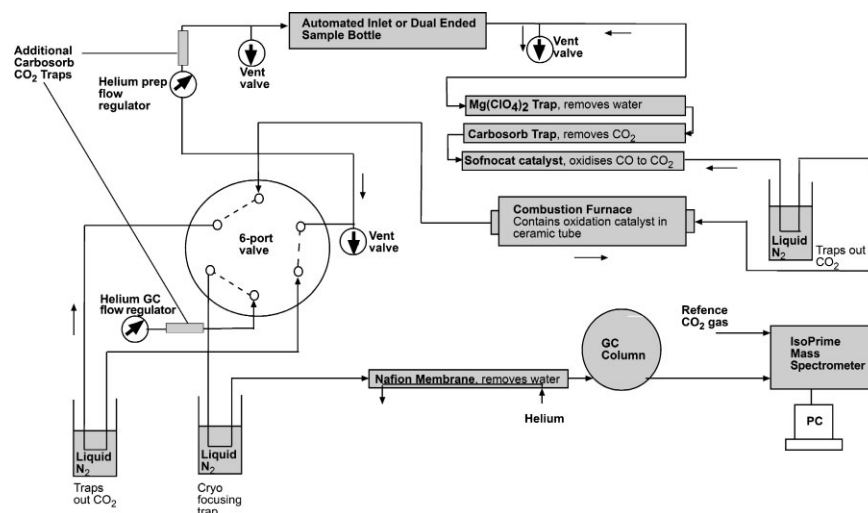


Figure 1. Schematic of the Trace Gas in the CH₄ analysis set-up. Black arrows indicate direction of flow of helium and the air sample during CH₄ analysis. Grey arrows show the default helium flow between analyses.

Quay *et al.*⁴ found background amplitudes to be 0.4‰ at 71°N, 0.3‰ at 48°N, 0.1‰ at 20°N and 14°S, and 0.2‰ between 25°S and 60°S. Measurement of the isotopic composition of background air at a precision that can resolve the seasonal cycle provides information on changing sources of the gases which can help refine global models. Background isotope measurements have been incorporated into some global models.^{13,14} For example, isotope measurements have proved useful in partitioning Siberian gas fields and Siberian wetlands, both of which are located in the same area but have isotopically distinct source signatures.¹³ An improved distribution of isotopic measurements could provide further improvements in global modelling and constraints on the global CH₄ budget.

On a regional scale, continental measurements become more useful. Continental background sites are often on mountains or tall towers, ideally a few hundred metres above the ground,¹⁵ such as the CHIOTTO network of tall towers in Europe.¹⁶ This allows measurements of well-mixed air, representative of the whole region. A mass balance approach, comparing the excess mixing ratio and isotopic composition over background levels, is used to identify the source signature of an area. Greater diurnal and day-to-day variability of continental site measurements arises because of the high variability in emission rates from sources that are predominantly terrestrial. For this reason, continuous measurements would be more useful in regional and local studies, but weekly flask sampling at marine background sites is likely to be adequate to pick up seasonal variations. Measurement of changing mixing ratios and stable isotope ratios during diurnal studies^{8,17} is a good method of determining sources of CH₄ and CO₂ on a regional and local scale. A record of $\delta^{13}\text{C}$ for CH₄ at a site over a period of time can provide important information about changing sources, and is a useful tool for verifying national emissions inventories.

This paper describes an automated inlet that has been developed with GV Instruments to analyse outside air at Royal Holloway (University of London), allowing measure-

ments of $\delta^{13}\text{C}$ of CH₄ or $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ of CO₂ to be made semi-continuously (normally at 30-min intervals) using the GV Instruments Trace Gas preparation system.

EXPERIMENTAL

Isotope measurement techniques

For analyses of $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ of CO₂ in air, water is first removed from the sample and the CO₂ is then cryotrapped at liquid nitrogen temperature while the remaining air in the sample is pumped away or vented. Precisions of 0.01‰ for $\delta^{13}\text{C}$ and 0.03‰ for $\delta^{18}\text{O}$ are routinely achieved using dual-inlet mass spectrometers.¹⁸ Isotopic analysis of $\delta^{13}\text{C}$ of CH₄ is more complicated. The most commonly used methods require combustion of CH₄ to CO₂ and water over a catalyst at >700°C, and subsequent isotopic analysis of the CO₂. The separation of CH₄ from other C-containing compounds in the air sample is achieved by one of two methods. Either the CH₄ is retained on a molecular sieve trap at liquid nitrogen temperature while allowing the other species in the air sample to pass through and then the CH₄ is desorbed prior to combustion,⁶ or chemical and cryogenic traps are used to remove the other C-containing compounds in the air sample.^{19,20} Measurements of $\delta^{13}\text{C}$ of CH₄ in air samples using these conventional off-line methods can require several tens to several hundreds of litres of air, and are labour intensive, but precisions of 0.02‰ have been achieved.²¹ High precisions of this order are required to resolve the small seasonal variations at low-latitude background sites.

Transportation costs, and the time taken to analyse the large-volume samples, limit the number of sites at which records can be taken if conventional techniques are used for CH₄ isotope analysis. High-precision isotopic analysis using small volumes of air is therefore highly advantageous, and allows a significant increase in the number of sites at which $\delta^{13}\text{C}$ of CH₄ can be measured.²² Commercially available preconcentrators linked to continuous-flow mass spectrometers are available to measure $\delta^{13}\text{C}$ of CH₄ (e.g. GV

Instruments Trace Gas, Thermofinnigan Precon,²³ Europa Anca TG-II), but the precision obtainable using these instruments off-the-shelf is currently not suitable for detecting annual changes at background stations. Work by several other groups^{22,24,25} has shown that precisions of better than 0.09‰ can be achieved for analysis of $\delta^{13}\text{C}$ of CH_4 in 40 to 200 mL samples. The preparation systems in these cases were laboratory-built, rather than representing minor modifications to a commercially available system as described in this work. The small-volume techniques are similar to each other in many ways. In the systems developed by Miller *et al.*,²² Rice *et al.*²⁵ and Morimoto *et al.*,²⁴ CH_4 is adsorbed on a Hayesep D packed column at -120°C as other air components are vented. The column is then heated and the CH_4 eluted and cryofocused. A gas chromatography (GC) column is used to separate components, CH_4 is oxidised to CO_2 in a combustion furnace, and finally the isotopic composition of the CH_4 is analysed by continuous-flow isotope-ratio mass spectrometry (CF-IRMS). CO_2 , carbon monoxide (CO) and other C-containing compounds are removed using chemical and liquid nitrogen traps in the GV Instruments Trace Gas device instead of isolating the CH_4 . Another difference is the positioning of the GC column, which in the Trace Gas is after the combustion furnace. This work describes minor modifications to the Trace Gas that have allowed $\delta^{13}\text{C}$ of CH_4 to be measured with a precision of 0.05‰ in 75 mL samples.

Standard Trace Gas instrumentation

The GV Instruments Trace Gas preconcentrator and IsoPrime mass spectrometer were installed in the Royal Holloway (RHUL) laboratory in March 2003. The instruments are connected in a continuous-flow set-up; a schematic of the arrangement for $\delta^{13}\text{C}$ of CH_4 analysis is shown in Fig. 1. Air samples enter the Trace Gas through a dual-ended 250–300 mm long sample bottle with two manually operated valves, or the newly developed automated inlet system. These are connected to the Trace Gas with two $1/4''$ Cajon Ultra-Torr fittings. A PC running Masslynx 4.0 software uses a GV Instruments defined analysis routine to control the valves on the Trace Gas. Whitey valves on the front panel of the system allow manual selection of the gas to be analysed (CH_4 , CO_2 or N_2O), and thus the route that the sample will follow through the Trace Gas. Each analysis takes 16 min.

During CH_4 analysis a ~ 20 mL/min flow of helium (99.999% purity, Air Products) transports the air sample through traps containing magnesium perchlorate (Elemental Microanalysis Ltd.) and Carbosorb (Elemental Microanalysis Ltd.) to remove water and CO_2 . Additional Carbosorb traps on the helium inlet lines have been added to completely ensure that there is no CO_2 present in the helium flow that would contaminate the sample. CO is oxidised to CO_2 by a Sofnocat catalyst (Molecular Products Ltd.), and the resultant CO_2 is then trapped in a liquid nitrogen cryotrap held at -196°C . The cryotrap is loops of nickel tube wrapped with glass-coated heater elements that automatically turn on when the traps are raised out of the liquid nitrogen. The flow continues to the combustion furnace where CH_4 is oxidised to CO_2 . The resultant CO_2 is trapped and cryofocused in the liquid nitrogen, and passes through a Nafion membrane to

remove any water produced during the oxidation step. The CO_2 is separated from any residual gas components in the sample in a 25 m PoraPLOT Q GC column (0.32 mm i.d.), and passes through an open split. The flow rates of the helium carrier gas are 1 mL/min at the head of the GC column and 0.7 mL/min at the open split. Part of the CO_2 is carried though to the IsoPrime mass spectrometer at a flow rate of 0.3 mL/min. The IsoPrime is tuned for isotopic analysis at m/z 44, 45 and 46. Measurements are made relative to a reference CO_2 gas cylinder (Air Products 4.5 grade).

During isotopic analysis of atmospheric CO_2 , the magnesium perchlorate trap removes water from the air sample, but the Carbosorb and Sofnocat traps and furnace are bypassed. A lower helium flow rate of 10 mL/min is used for CO_2 analysis.

Calibration

The IsoPrime mass spectrometer has been calibrated for measurement of CO_2 derived from greenhouse gases with a range of $\delta^{13}\text{C}$ from -8 to -80 ‰, using the following procedures. First, the CO_2 reference gas cylinder was calibrated against the reference gas of a dual inlet PRISM (VG Isotech) mass spectrometer at RHUL. The pure CO_2 working reference of the PRISM is calibrated against IAEA (International Atomic Energy Agency) carbonate standards,²⁶ NBS-19 and CO-9, over a $\delta^{13}\text{C}_{\text{VPDB}}$ range from -47 to $+2$ ‰ and $\delta^{18}\text{O}_{\text{VPDB-CO}_2}$ from -25 to -12 ‰. Second, the Trace Gas has been calibrated for analysis of CO_2 in air against a primary air calibration standard purchased from the National Oceanic and Atmospheric Administration (NOAA), with $\delta^{13}\text{C}_{\text{VPDB}} -8.431 \pm 0.002$ ‰ and $\delta^{18}\text{O}_{\text{VPDB-CO}_2} -3.728 \pm 0.013$ ‰. Third, the magnitude of the mass spectrometer scaling effect was determined by analysis with the Trace Gas of $\delta^{13}\text{C}$ of CO_2 in a light CO_2 standard (Heidi Light at -50.17 ‰) from IUP (Institut für Umweltphysik, Heidelberg), diluted with zero air to ambient concentration. Trace Gas measurements were 0.80‰ higher than the true value. Finally, the system (as well as a conventional CH_4 extraction system at RHUL²⁰) was calibrated for $\delta^{13}\text{C}$ of CH_4 in the -50 to -47 ‰ range as part of a round-robin exercise for Meth-MonitEUr (an EU-funded methane monitoring consortium). Two air standards were measured, NZ1 and EG96, for which conventional, dual-inlet analyses gave values of -47.21 ± 0.03 ‰ and -50.40 ± 0.04 ‰, respectively. Additionally, an internal working standard (RHS 584, containing 560 L of air, $\delta^{13}\text{C}$ value of -47.48 ± 0.03 ‰) was measured using the conventional CH_4 extraction line at RHUL (long-term precision of ± 0.04 ‰) and then analysed routinely using the Trace Gas. This enables corrections to be made to sample results if there is a drift in $\delta^{13}\text{C}$ of CH_4 of the secondary standard. The raw values of $\delta^{13}\text{C}$ are offset by 0.8‰ which can be attributed to the mass spectrometer scaling effect. Therefore, there is no significant offset resulting from the process of oxidation of CH_4 to CO_2 (see following discussion).

Oxidation catalyst

The efficiency of the conversion of CH_4 into CO_2 in the furnace is a major limitation to the precision that can be obtained by the GC/IRMS method. Use of different catalysts, furnace temperatures and flow rates in the Trace Gas have

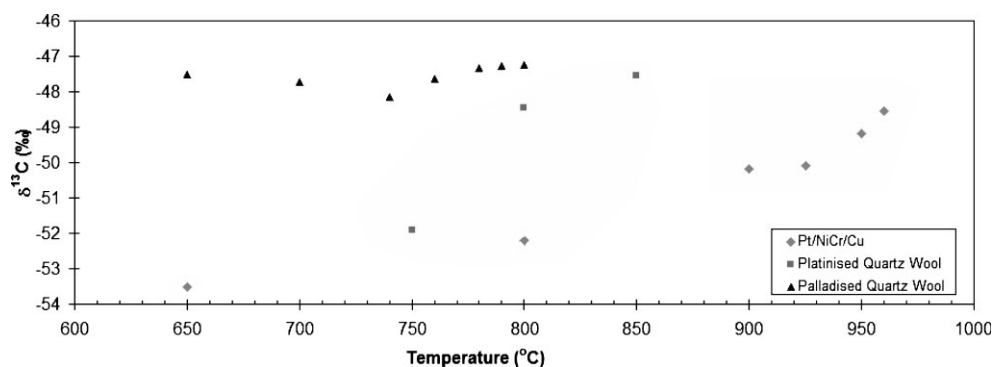


Figure 2. Comparison of $\delta^{13}\text{C}$ of CH_4 for RHS 584 (averaged over triplicate measurements) against temperature for the three catalysts.

been found to have a large effect on the efficiency of oxidation and hence the $\delta^{13}\text{C}$ value, the height of the CO_2 peak measured by the mass spectrometer, as well as the precision.

Three oxidation catalysts have been investigated: braided wires of platinum, nichrome and copper (Pt/NiCr/Cu); platinised quartz wool; and palladised quartz wool. Each catalyst was tested in the furnace at a range of temperatures, from 650°C upwards, with the aim of achieving the highest possible repeatability and accuracy. Repeatability, i.e. precision, was defined as the single standard deviation for ten consecutive analyses of the internal secondary standard, RHS 584. Samples were introduced into the Trace Gas via a 150 mL stainless steel sample volume during the catalyst tests. The sample volume was connected to a pump and the secondary standard tank via two manually operated three-way valves, enabling the sample volume to be evacuated and filled from the tank without being disconnected from the Trace Gas. This was a manual version of the design later automated.

On installation, the Trace Gas furnace contained braided 0.125 mm diameter platinum, nichrome and copper wires (Advent) in a ceramic tube (99.7% Al_2O_3) of i.d. 0.5 mm and length 450 mm. A temperature of 900°C and helium flow rate of 20 mL/min were initially used. The copper was oxidised for 30 min at the start of each day using a flow rate of 2 mL/min mixed with the helium flow. Without oxidation, $\delta^{13}\text{C}$ values were lighter than expected, indicating inefficient CH_4 oxidation in the furnace. With regular daily oxidation there was still a trend in $\delta^{13}\text{C}$ values for a standard tank throughout the day. A full day of standard measurements shows values depleting by nearly 2‰ over the course of 17 measurements. Although this can to some extent be corrected for by analysing a standard tank at intervals throughout the day, and using this to correct the other measurements, the corrected results would not reach the required precision. Use of the fixed inlet system, rather than individual bottles connected to the Trace Gas, improved the precision obtained with the original catalyst from 0.25 to 0.19‰. As well as $\delta^{13}\text{C}$ values, peak heights can be used to give an indication of yield, and hence the oxidation efficiency. The heaviest values and highest precisions and peak heights were measured at a temperature of 960°C and helium flow rate of 17 mL/min. Higher flow rates led to a depletion in the measured $\delta^{13}\text{C}$ values as well as reducing peak height. This is indicative of partial oxidation. Higher flow rates minimise the amount of time during which the CH_4 sample remains in the furnace

and there is not enough time for complete oxidation of the sample to occur. As $^{12}\text{CH}_4$ is preferentially oxidised, incomplete oxidation will result in the CO_2 derived from CH_4 combustion being more depleted in ^{13}C than the true value.

Two grams of platinised quartz wool (Shimadzu high sensitivity catalyst), inserted in a ceramic tube of 4 mm i.d. (Ceramic Substrates & Components Ltd.), was tested in the furnace. At 800°C, peak heights and precision were at a maximum. The highest peak heights obtained using platinised quartz wool were 80% of the largest peaks obtained with the Pt/NiCr/Cu catalyst. It is possible that a higher density of platinum powder on the quartz wool (not specified by the manufacturer) would improve the yield, but this has not been tested. Oxidation of the catalyst was not required as oxidation of CH_4 to CO_2 used the oxygen present in the air sample, so the depletion in measured values throughout the day was not present with the platinised quartz wool. Hence, precision was improved compared with the original catalyst (from 0.19 to 0.14‰). In the Pt/NiCr/Cu catalyst, copper acts as a substrate for oxygen, and catalysts that utilise copper usually require conditioning with a flow of oxygen prior to use. The greater catalyst surface area with the platinised quartz wool also enables more of the oxygen from the air sample to reach a reactive interface and therefore be available for the oxidation reaction. As with the original catalyst, the highest peak heights were obtained with a helium flow rate of 17 mL/min.

Palladised quartz wool was made up by agitating 5 g of palladium powder (99.99% purity, Aldrich Chemical Co.) with 10 g of quartz wool until the product was a uniform grey colour.²⁷ Approximately 2 g of the palladised quartz wool was inserted in a 4 mm i.d. ceramic tube, filling a length of 25 cm. A length of 1 cm of pure quartz wool was inserted in each end of the ceramic tube to ensure that the fine palladium powder did not escape from the furnace tube and inhibit the gas flow. As with the previous platinised catalyst, the palladised quartz wool used oxygen present in the air sample during CH_4 combustion.

Figure 2 shows the $\delta^{13}\text{C}$ values for temperature tests using the three catalysts. Measurements at each temperature were made in triplicate on the secondary standard tank (RHS 584). Greatest stability is seen in the palladised quartz wool results. Above a temperature of 790°C the palladised quartz wool curve reaches a maximum value, closest to the conventional

Table 1. Optimum furnace temperature and the $\delta^{13}\text{C}$ values, peak heights and precision at this temperature for the catalysts tested in the Trace Gas. Precisions listed are defined as one standard deviation in ten consecutive analyses of a secondary standard tank (RHS 584) using a fixed inlet system and a 150 mL sample volume. The RHUL conventional extraction line measurement of RHS 584 was $-47.48 \pm 0.03\%$

Catalyst	Optimum temperature ($^{\circ}\text{C}$)	$\delta^{13}\text{C}$ (‰)	Peak height (nA)	Precision in 10 analyses (‰)
Braided Pt, Cu & NiCr wires	960	-48.76	3.0	0.19
Platinised quartz wool	800	-48.66	2.4	0.14
Palladised quartz wool	790	-47.48	7.1	0.09

extraction line value for RHS 584 of -47.48% . This indicates that, above 790°C , the CH_4 is completely oxidised. The other catalysts showed a much greater variation with temperature, and, at each temperature, $\delta^{13}\text{C}$ values were more depleted than with the palladised quartz wool, indicating that incomplete combustion occurred.

Table 1 summarises the catalyst tests. The best precision (0.09‰) and highest peak heights were obtained with the palladised quartz wool catalyst at a temperature of 790°C . The greater than twofold increase in peak height of CO_2 generated from palladium versus Pt/NiCr/Cu wire catalyst, and the enriched $\delta^{13}\text{C}$ values which are within 1σ error of the expected value, suggest that there is greater combustion efficiency with the palladium. Thus, use of palladium as a catalyst improves accuracy, precision and yield.

Optimum flow (denoted by maximum peak heights) occurred above 40 mL/min for the palladised quartz wool catalyst at 790°C (Fig. 3) for both 75 and 150 mL sample volumes, i.e. double the flow rate used with the other catalysts. This suggests that sample removal from the sample volumes is efficient. The variation in optimum flow rate is likely to be highly dependent on the surface area of the active catalyst, i.e. how densely the furnace tube is packed.

After addition of the automated inlet (described in the following section), precision was further improved. CO_2 repeatability of 0.03% for $\delta^{13}\text{C}$ and 0.05% for $\delta^{18}\text{O}$, and CH_4 repeatability of 0.05% , for ten consecutive analyses of RHS 584, are routinely achieved. Tests on the Pt/Cu/NiCr wire and platinised quartz wool catalysts were discontinued prior

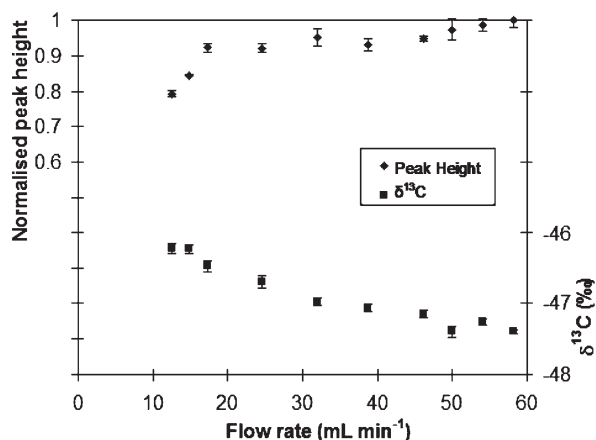


Figure 3. Peak height and corrected methane $\delta^{13}\text{C}$ variation with flow rate using the palladised quartz wool catalyst at a furnace temperature of 790°C . Values are averages of measurements of the NOAA 3 tank using a 75 mL sample volume.

to the installation of the automated inlet in November 2004 to focus on the more efficient palladised quartz wool. Figure 4 lists the corrected $\delta^{13}\text{C}$ value and peak height for triplicate analyses of the NOAA 3 tank made with the palladised quartz wool catalyst using the automated inlet. For comparison, the RHUL conventional extraction line $\delta^{13}\text{C}$ value for NOAA 3 was -47.20% . Above 790°C the $\delta^{13}\text{C}$ and peak heights reach a maximum and stabilise, with best precision occurring at 790°C . As a result of the findings of these catalyst tests, palladised quartz wool at a temperature of 790°C has been used as the catalyst for all subsequent analyses with a flow rate of 40 mL/min .

Automated inlet system

In a collaboration between RHUL and GV Instruments, an automated inlet system has been developed. A schematic is shown in Fig. 5. Two of the ports on a six-port $1/8''$ Valco valve are connected to a 75 mL stainless steel volume (Swagelok) and two ports are connected to the $1/4''$ Cajon Ultra-Torr fittings on the Trace Gas. The other two ports are connected to T-piece fittings and then to four shut-off valves (V1, V2, V3 and V4). Valve V1 is connected to a rotary pump (Edwards) and Pirani gauge (Edwards), to evacuate the sample volume between analyses before it is filled with air from a tank. V3 is attached to a bleed valve and a diaphragm pump (KNF Neuberger) which pumps air continuously from an inlet on the roof of the building, approximately 15 m above ground level. V2 is used as a vent valve that is opened when outside air is flushing through the sample volume. V4 is connected to a 70 L tank containing an internal secondary

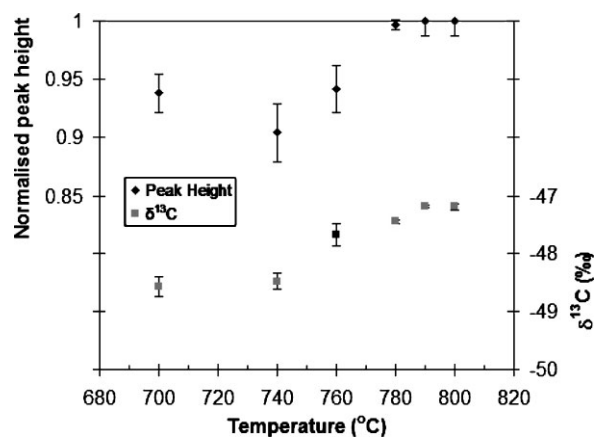


Figure 4. Corrected $\delta^{13}\text{C}$ values and peak heights for triplicate analyses of NOAA 3 in a 75 mL volume with the palladised quartz wool catalyst.

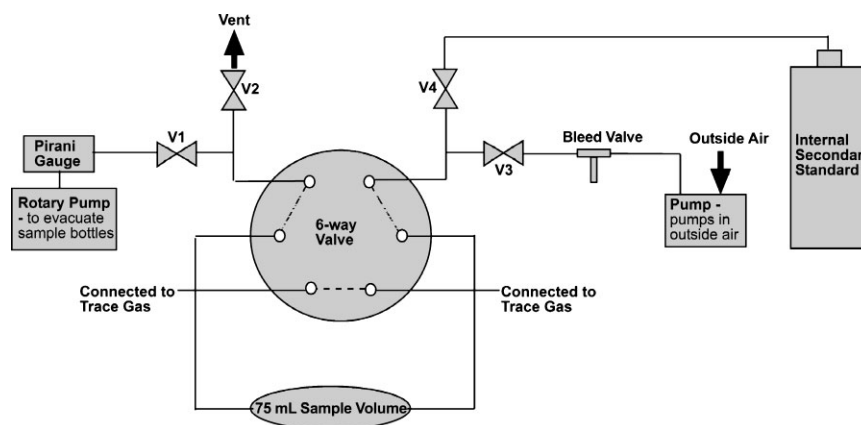


Figure 5. Schematic of the automated inlet system in load position.

standard. All connecting tubing is $\frac{1}{8}$ " stainless steel. Each of these valves is air actuated using a 65 psi flow of compressed air. The six-port valve switches to allow evacuation/filling/flushing of the sample volume when in the load position, or directs flow of the sample from the sample volume to the Trace Gas when in the inject position. The inlet is controlled by the MassLynx 4.0 software that controls the Trace Gas and IsoPrime. This inlet is now an optional extra when purchasing a Trace Gas preconcentrator from GV Instruments.

On commencement of an analysis sequence, the six-port valve is set to the load position. Opening V1 for 90 s, while V2, V3 and V4 are closed, evacuates the sample volume to a pressure of 1×10^{-2} mbar. The sample volume does

not have stopcocks, but uses the two-way valves to trap samples within it. The sample volume can then be filled with air from the secondary standard tank by closing V1 and opening V4 for 30 s. Thus the sample volume is vacuum-filled when a tank is analysed to a maximum of 4 psi above ambient pressure. Alternatively, when filling the sample volume with contemporaneous outside air, V3 and V2 are opened for at least 120 s (while V1 and V4 are closed), and the air flushes through the sample volume and vents until V3 and V2 are closed. Whenever the six-port valve is in the load position, helium passes directly through the Trace Gas bypassing the sample volume. Once all the valves (V1–V4) are closed and the sample volume has been filled, the six-port valve is switched to the inject position and helium

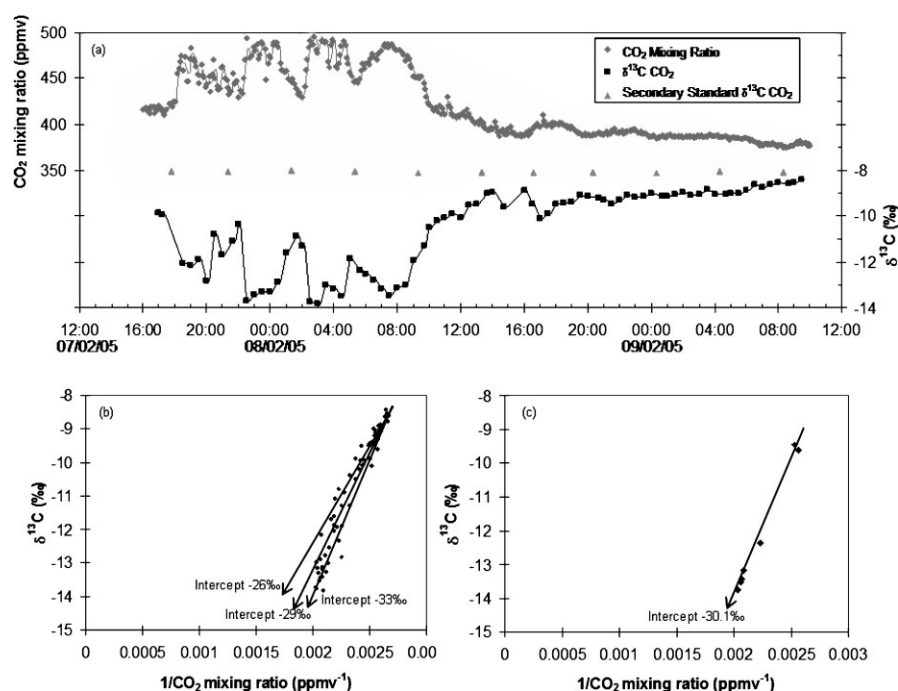


Figure 6. CO₂ diurnal study, February 7–9, 2005. (a) Time series of CO₂ mixing ratios and δ¹³C of CO₂. Secondary standard tank δ¹³C of CO₂ analyses are also shown. (b) Keeling plot for all wind directions during the diurnal. (c) Keeling plot for easterly wind directions. The best approximation of the source isotopic signature is indicated where the line crosses the y-axis, at 1/CO₂ = 0.

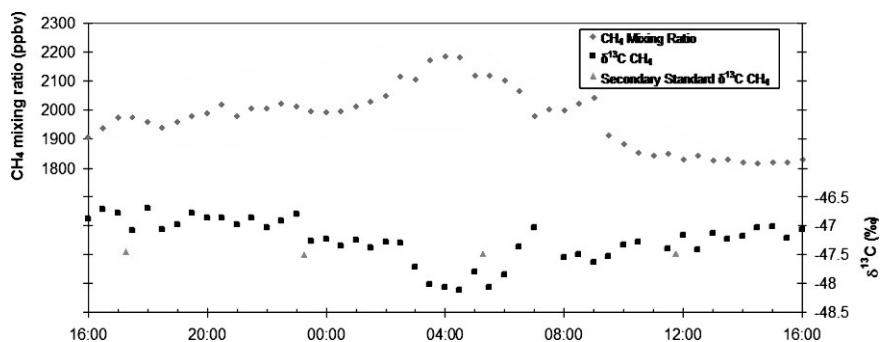


Figure 7. Methane diurnal study 27–28 July, 2005 (wind direction E): time series of CH_4 mixing ratio and $\delta^{13}\text{C}$ of CH_4 . Secondary standard tank analyses of $\delta^{13}\text{C}$ are also shown.

flows through the sample volume before passing into the Trace Gas.

When the outside air inlet is not required, three tanks, flasks or sample bags can be connected to V2, V3 and V4, allowing automated analysis of a set of samples. The sample volume is then filled using the vacuum filling procedure.

Diurnal studies using the Trace Gas and automated inlet system

The Royal Holloway (RHUL) sampling site is situated in Egham, Surrey, on the first significant incline west of London ($51^\circ 25.6'\text{N}$, $0^\circ 33.7'\text{W}$), 32 km from the centre of the city. It is well situated to measure air that has passed over London when winds are easterly. During still, high-pressure, anticyclonic conditions, an inversion builds up over the London basin and there is a build-up of pollutants emitted by the city. If there are gentle easterly winds the build-up of carbon gases measured at RHUL can be significant, with excesses over background of >1000 ppbv of CH_4 and 50 ppmv of CO_2 frequently measured. Measurement of the stable C isotopic composition of these gases can provide useful information about the major sources of the gases in the London area. In previous years the diurnal sampling campaigns at RHUL have involved collecting air in 22 L tanks pressurised to 8 bars at 2-h intervals throughout the night.^{20,28} These samples were then analysed on the RHUL cryogenic extraction line at a later date. The automated inlet on the Trace Gas now makes it possible to conduct a greater number of diurnal studies with measurements made more frequently.

Currently, the Trace Gas at RHUL is run overnight for either CO_2 or CH_4 isotope studies on occasions when the meteorology is such that there is likely to be a large build-up of carbon gases. When diurnal studies are performed, outside air is analysed every 30 min and the secondary standard is measured every 4 or 6 h. Isotopic values are compared with CO_2 mixing ratios recorded at 5-min intervals on a LICOR 6252 CO_2 analyser (± 0.3 ppmv), and CH_4 mixing ratios are measured using a HP5890 GC (± 3 ppbv) at 30-min intervals. Meteorological information (wind direction, wind speed, pressure, relative humidity and temperature) is also recorded at the site.

Examples of the results obtained in diurnal studies are shown in Figs. 6(a) and 7. The results presented here are from preliminary studies, and are provided as an example of how

GC/IRMS can produce continuous records of CH_4 or CO_2 isotopic variations with a precision that is useful for regional and local studies of emissions. The precision (reproducibility, 1σ) for $\delta^{13}\text{C}$ in 11 analyses of the standard tank over the 40-h time period of the CO_2 experiment was 0.04‰ (Fig. 6(a)). The wind was mainly from the east (i.e. London) during the first night, but there were slight changes in direction (from ENE to ESE). Four distinct peaks overnight represent changing wind directions. The wind was predominantly from the SW during the second half of the period and there was no overnight build-up during the second night. By subdividing the data according to wind direction, source values for different sectors can be calculated. Source signatures are calculated using the 'Lever' rule:²⁹

$$\delta^{13}\text{C}_{\text{source}} = (\delta^{13}\text{C}_{\text{air}} \times c_{\text{air}} - \delta^{13}\text{C}_{\text{background}} \times c_{\text{background}}) / \Delta c$$

where Δc is the difference in mixing ratio c between the selected background and the sampled air (c_{air}). $\delta^{13}\text{C}_{\text{air}}$ differs from $\delta^{13}\text{C}_{\text{background}}$ depending on the isotopic signature of the CH_4 from emission sources that is mixed with background air. For the easterly sector (Fig. 6(c)), i.e. air that has come from London, the overnight source of excess CO_2 had an isotopic composition of -30.1‰ . For comparison, typical values of $\delta^{13}\text{C}$ of CO_2 are -24‰ for coal combustion, -40‰ from thermogenic natural gas, -27‰ from vehicle emissions, and -26‰ from biogenic respiration, although there can be significant geographical variation in these values.^{9,30}

During the CH_4 diurnal study shown in Fig. 7, the wind was predominantly from the easterly sector which encompasses the London region. The mean overnight source mix (for the period 22:00 until 06:00), calculated using the Lever rule, was $-50.5 \pm 0.7\text{‰}$. Sources of CH_4 in this region are dominated by isotopically depleted landfill emissions (typically -53‰) and more enriched gas distribution losses.^{20,28} The source mix calculated from this diurnal study can be apportioned to 82% landfill and 18% gas distribution losses, which agrees closely with estimates obtained from earlier diurnal studies conducted at RHUL using conventional methods of CH_4 isotope analysis.²⁰ Measurements of $\delta^{13}\text{C}$ of CH_4 for the secondary standard tank had a reproducibility of 0.05‰ in five analyses over 24 h, and there was no noticeable catalyst memory effect when switching between outside air with a $\delta^{13}\text{C}$ spread of 1.5‰ and the secondary standard tank.

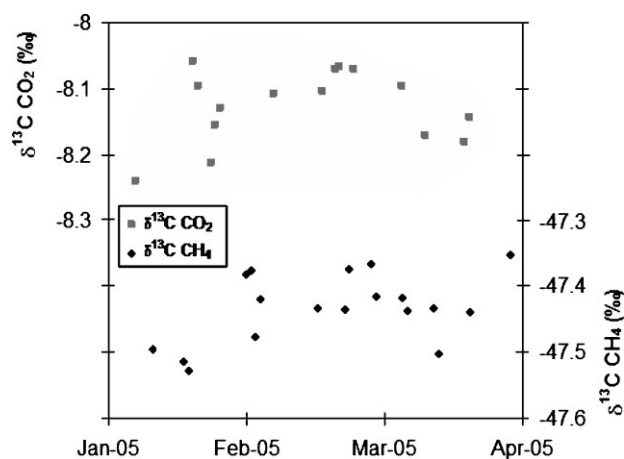


Figure 8. Daily averaged $\delta^{13}\text{C}$ of CO_2 and $\delta^{13}\text{C}$ of CH_4 values for RHS 584 tank analyses over a 3-month period. Variability for CO_2 is of the same order of magnitude as for CH_4 indicating that variability is not related to the CH_4 combustion step.

DISCUSSION

Palladium-based catalysts are commonly used for oxidation of CH_4 , e.g. in natural gas fuelled turbines or in natural gas vehicles,³¹ but the use of palladium powder on quartz wool for complete CH_4 oxidation in GC/IRMS is new. The good repeatability achieved, and the increased peak heights, indicate that palladised quartz wool was the most efficient of the three catalysts tested. The main problem with palladised quartz wool was that it took more than 2 weeks of running standards until the furnace was conditioned and the results stabilised. In contrast, conditioning the Pt/Cu/NiCr catalyst only took 1 day. Analysis of helium blanks yielded large CO_2 peaks when the palladised quartz wool furnace tube was first installed. These blank peaks were >10 nA on the first day of the new catalyst and were reduced to <0.3 nA after eight days of analyses using the 150 mL sample volume. A much larger effect was seen with the platinised quartz wool catalyst, with blank peaks of 0.7 nA still present after 8 weeks. This was probably caused by residual carbon or C-containing compounds being burnt off. The conditioning period could be reduced by flushing oxygen through the furnace tube at reaction temperature, or conditioning the quartz wool prior to installation of each new furnace tube. With the automated inlet that is now in operation, the conditioning time could be greatly reduced by continuous overnight analysis of aliquots of zero air.

The system has now run for more than a year with the same catalyst with no noticeable deterioration in its efficiency. Daily averages of $\delta^{13}\text{C}$ measurements of the standard tank (RHS 584) for CO_2 and CH_4 over a period of 3 months, following installation of the automated valves, are shown in Fig. 8. The precision (1σ) over this period was 0.05‰ for $\delta^{13}\text{C}$ of both CH_4 and CO_2 . Over a longer time period the precision decreases slightly, but variations in $\delta^{13}\text{C}$ of CH_4 over time closely match variations in $\delta^{13}\text{C}$ of CO_2 , and thus are not caused by perturbations in oxidation efficiency in the furnace. Most deviations in standard values

can be related to power outages or prolonged periods of machine inactivity.

Benefits of the automated valve system

Addition of the automated inlet system yielded a further substantial improvement in the precision and accuracy of results. This is partly due to a reduction in contamination with laboratory air through the connections to the sample volume, which can occur when individual sample bottles are connected to the Trace Gas. Another reason is the elimination of variations in pump time, fill time and time between analyses that were present when valves were opened manually and each sample run had to be individually initialised. As well as the convenience of being able to run a set of samples back-to-back without manually opening valves, it has enabled a quicker throughput of samples and, more importantly, an increased number of diurnal studies to be conducted.

Currently, the RHUL system uses a 3-L Dewar which requires the liquid nitrogen level to be topped up every 8 h. With the additional expense of an automated liquid nitrogen dispenser the system could be run continuously, so a continuous record of 30-min measurements of $\delta^{13}\text{C}$ of CH_4 (or $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ of CO_2) could be obtained. A record of this at a site over a decadal period would provide information about variations in sources of these important greenhouse gases.

CONCLUSIONS

The results discussed in this paper have demonstrated that precisions of 0.03‰ for $\delta^{13}\text{C}$ and 0.05‰ for $\delta^{18}\text{O}$ of atmospheric carbon dioxide, and 0.05‰ for $\delta^{13}\text{C}$ of atmospheric methane can be achieved with CF-IRMS, by making small-scale modifications to the GV Instruments Trace Gas system. Repeatability was improved with the use of a catalyst consisting of palladium powder on quartz wool for CH_4 combustion and by using a fixed and automated inlet system. Whereas CF-IRMS has in the past found its niche in source studies, this precision makes the instrument suitable for measuring diurnal variations at urban sites and has great potential for measuring seasonal variations at background stations. The small sample volume, rapid analysis time and high precision will enable a significant increase in the number of CH_4 and CO_2 isotope measurements in air samples to be made, providing additional data for developing global models of isotopic change, an important step to understanding changing emissions and verification of emissions inventories. By running the system continuously, changes in the magnitude and type of sources of these important greenhouse gases could be recognised both seasonally and inter-annually.

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